

11.13 Exercises

Exercise 11.1 Classify the following robot tasks as motion control, force control, hybrid motion–force control, impedance control, or some combination. Justify your answer.

- (a) Tightening a screw with a screwdriver.
- (b) Pushing a box along the floor.
- (c) Pouring a glass of water.
- (d) Shaking hands with a human.
- (e) Throwing a baseball to hit a target.
- (f) Shoveling snow.
- (g) Digging a hole.
- (h) Giving a back massage.
- (i) Vacuuming the floor.
- (j) Carrying a tray of glasses.

Exercise 11.2 The 2% settling time of an underdamped second-order system is approximately $t = 4/(\zeta\omega_n)$, for $e^{-\zeta\omega_n t} = 0.02$. What is the 5% settling time?

Exercise 11.3 Solve for any constants and give the specific equation for an underdamped second-order system with $\omega_n = 4$, $\zeta = 0.2$, $\theta_e(0) = 1$, and $\dot{\theta}_e(0) = 0$. Calculate the damped natural frequency, approximate overshoot, and 2% settling time. Plot the solution on a computer and measure the exact overshoot and settling time.

Exercise 11.4 Solve for any constants and give the specific equation for an underdamped second-order system with $\omega_n = 10$, $\zeta = 0.1$, $\theta_e(0) = 0$, and $\dot{\theta}_e(0) = 1$. Calculate the damped natural frequency. Plot the solution on a computer.

Exercise 11.5 Consider a pendulum in a gravitational field with $g = 10 \text{ m/s}^2$. The pendulum consists of a 2 kg mass at the end of a 1 m massless rod. The pendulum joint has a viscous-friction coefficient of $b = 0.1 \text{ N m s/rad}$.

- (a) Write the equation of motion of the pendulum in terms of θ , where $\theta = 0$ corresponds to the “hanging down” configuration.
- (b) Linearize the equation of motion about the stable “hanging down” equilibrium. To do this, replace any trigonometric terms in θ with the linear term in the Taylor expansion. Give the effective mass and spring con-

stants m and k in the linearized dynamics $\mathbf{m}\ddot{\theta} + b\dot{\theta} + k\theta = 0$. At the stable equilibrium, what is the damping ratio? Is the system underdamped, critically damped, or overdamped? If it is underdamped, what is the damped natural frequency? What is the time constant of convergence to the equilibrium and the 2% settling time?

- (c) Now write the linearized equations of motion for $\theta = 0$ at the balanced upright configuration. What is the effective spring constant k ?
- (d) You add a motor at the joint of the pendulum to stabilize the upright position, and you choose a P controller $\tau = K_p\theta$. For what values of K_p is the upright position stable?

Exercise 11.6 Develop a controller for a one-dof mass-spring-damper system of the form $\mathbf{m}\ddot{x} + b\dot{x} + kx = f$, where f is the control force and $\mathbf{m} = 4$ kg, $b = 2$ Ns/m, and $k = 0.1$ N/m.

- (a) What is the damping ratio of the uncontrolled system? Is the uncontrolled system overdamped, underdamped, or critically damped? If it is underdamped, what is the damped natural frequency? What is the time constant of convergence to the origin?
- (b) Choose a P controller $f = K_px_e$, where $x_e = x_d - x$ is the position error and $x_d = 0$. What value of K_p yields critical damping?
- (c) Choose a D controller $f = K_d\dot{x}_e$, where $\dot{x}_d = 0$. What value of K_d yields critical damping?
- (d) Choose a PD controller that yields critical damping and a 2% settling time of 0.01 s.
- (e) For the PD controller above, if $x_d = 1$ and $\dot{x}_d = \ddot{x}_d = 0$, what is the steady-state error $x_e(t)$ as t goes to infinity? What is the steady-state control force?
- (f) Now insert a PID controller for f . Assume $x_d \neq 0$ and $\dot{x}_d = \ddot{x}_d = 0$. Write the error dynamics in terms of $\ddot{x}_e$, $\dot{x}_e$, x_e , and $\int x_e(t)dt$ on the left-hand side and a constant forcing term on the right-hand side. (Hint: You can write kx as $-k(x_d - x) + kx_d$.) Take the time derivative of this equation and give the conditions on K_p , K_i , and K_d for stability. Show that zero steady-state error is possible with a PID controller.

Exercise 11.7 Simulation of a one-dof robot and robot controller.

- (a) Write a simulator for a one-joint robot consisting of a motor rotating a link in gravity using the model parameters given in Section 11.4.1. The simulator should consist of: (1) a dynamics function that takes as input the current state of the robot and the torque applied by the motor and gives as output the acceleration of the robot; and (2) a numerical integrator that

uses the dynamics function to calculate the new state of the system over a series of timesteps Δt . A first-order Euler integration method suffices for this problem (e.g., $\theta(k+1) = \theta(k) + \dot{\theta}(k)\Delta t$, $\dot{\theta}(k+1) = \dot{\theta}(k) + \ddot{\theta}(k)\Delta t$). Test the simulator in two ways: (1) starting the robot at rest at $\theta = -\pi/2$ and applying a constant torque of 0.5 N m; and (2) starting the robot at rest at $\theta = -\pi/4$ and applying zero torque. For both examples, plot the position as a function of time for sufficient duration to see the basic behavior. Ensure that the behavior makes sense. A reasonable choice of Δt is 1 ms.

- (b) Add two more functions to your simulator: (1) a trajectory generator function that takes the current time and returns the desired state and acceleration of the robot; and (2) a control function that takes the current state of the robot and information from the trajectory generator and returns a control torque. The simplest trajectory generator would return $\theta = \theta_{d1}$ and $\dot{\theta} = \dot{\theta} = 0$ for all time $t < T$, and $\theta = \theta_{d2} \neq \theta_{d1}$ and $\dot{\theta} = \dot{\theta} = 0$ for all time $t \geq T$. This trajectory is a step function in position. Use a PD feedback controller for the control function and set $K_p = 10$ N m/rad. For a well-tuned choice of K_d , give K_d (including units) and plot the position as a function of time over 2 seconds for an initial state at rest at $\theta = -\pi/2$ and a step trajectory with $\theta_{d1} = -\pi/2$ and $\theta_{d2} = 0$. The step occurs at $T = 1$ s.
- (c) Demonstrate two different choices of K_d that yield (1) overshoot and (2) sluggish response with no overshoot. Give the gains and the position plots.
- (d) Add a nonzero K_i to your original well-tuned PD controller to eliminate steady-state error. Give the PID gains and plot the results of the step test.

Exercise 11.8 Modify the simulation of the one-joint robot in Exercise 11.7 to model a flexible transmission from the motor to the link with a stiffness of 500 Nm/rad. Tune a PID controller to give a good response for a desired trajectory with a step transition from $\theta = -\pi/2$ to $\theta = 0$. Give the gains and plot the response.

Exercise 11.9 Simulation of a two-dof robot and robot controller (Figure 11.25).

- (a) *Dynamics.* Derive the dynamics of a 2R robot under gravity (Figure 11.25). The mass of link i is m_i , the center of mass is a distance r_i from the joint, the scalar inertia of link i about the joint is $\mathcal{I}_i$, and the length of link i is L_i . There is no friction at the joints.
- (b) *Direct drive.* Assume each joint is directly driven by a DC motor with

no gearing. Each motor comes with specifications of the mass $\mathbf{m}_i^{\text{stator}}$ and inertia $\mathcal{I}_i^{\text{stator}}$ of the stator and the mass $\mathbf{m}_i^{\text{rotor}}$ and inertia $\mathcal{I}_i^{\text{rotor}}$ of the rotor (the spinning portion). For the motor at joint i , the stator is attached to link $i - 1$ and the rotor is attached to link i . The links are thin uniform-density rods of mass $\mathbf{m}_i$ and length L_i .

In terms of the quantities given above, for each link $i \in \{1, 2\}$ give equations for the total inertia $\mathcal{I}_i$ about the joint, the mass $\mathbf{m}_i$, and the distance r_i from the joint to the center of mass. Think about how to assign the mass and inertia of the motors to the different links.

- (c) *Geared robot.* Assume that motor i has gearing with gear ratio G_i and that the gearing itself is massless. As in part (b) above, for each link $i \in \{1, 2\}$, give equations for the total inertia $\mathcal{I}_i$ about the joint, mass $\mathbf{m}_i$, and distance r_i from the joint to the center of mass.
- (d) *Simulation and control.* As in Exercise 11.7, write a simulator with (at least) four functions: a dynamics function, a numerical integrator, a trajectory generator, and a controller. Assume that there is zero friction at the joints, $g = 9.81 \text{ m/s}^2$ in the direction indicated, $L_i = 1 \text{ m}$, $r_i = 0.5 \text{ m}$, $\mathbf{m}_1 = 3 \text{ kg}$, $\mathbf{m}_2 = 2 \text{ kg}$, $\mathcal{I}_1 = 2 \text{ kg m}^2$, and $\mathcal{I}_2 = 1 \text{ kg m}^2$. Program a PID controller, find gains that give a good response, and plot the joint angles as a function of time for a reference trajectory which is constant at $(\theta_1, \theta_2) = (-\pi/2, 0)$ for $t < 1 \text{ s}$ and constant at $(\theta_1, \theta_2) = (0, -\pi/2)$ for $t \geq 1 \text{ s}$. The initial state of the robot is at rest with $(\theta_1, \theta_2) = (-\pi/2, 0)$.
- (e) *Torque limits.* Real motors have limits on the available torque. While these limits are generally velocity dependent, here we assume that each motor's torque limit is independent of velocity, $\tau_i \leq |\tau_i^{\text{max}}|$. Assume that $\tau_1^{\text{max}} = 100 \text{ N m}$ and $\tau_2^{\text{max}} = 20 \text{ N m}$. The control law may request greater torque but the actual torque is saturated at these values. Rerun the PID control simulation in (d) and plot the torques as well as the position as a function of time.
- (f) *Friction.* Add a viscous friction coefficient of $b_i = 1 \text{ N m s/rad}$ to each joint and rerun the PID control simulation in (e).

Exercise 11.10 For the two-joint robot of Exercise 11.9, write a more sophisticated trajectory generator function. The trajectory generator should take the following as input:

- the desired initial position, velocity, and acceleration of each joint;
- the desired final position, velocity, and acceleration; and
- the total time of motion T .

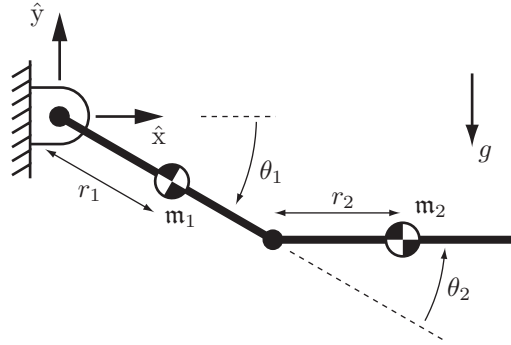


Figure 11.25: A two-link robot arm. The length of link i is L_i and its inertia about the joint is $\mathcal{I}_i$. The acceleration due to gravity is $g = 9.81 \text{ m/s}^2$.

A call of the form

```
[qd,qdtd,qdtdtd] = trajectory(time)
```

returns the desired position, velocity, and acceleration of each joint at time `time`. The trajectory generator should provide a trajectory that is a smooth function of time.

As an example, each joint could follow a fifth-order polynomial trajectory of the form

$$\theta_d(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3 + a_4 t^4 + a_5 t^5. \quad (11.67)$$

Given the desired positions, velocities, and accelerations of the joints at times $t = 0$ and $t = T$, you can uniquely solve for the six coefficients $a_0, \dots, a_5$ by evaluating Equation (11.67) and its first and second derivatives at $t = 0$ and $t = T$.

Tune a PID controller to track a fifth-order polynomial trajectory moving from rest at $(\theta_1, \theta_2) = (-\pi/2, 0)$ to rest at $(\theta_1, \theta_2) = (0, -\pi/2)$ in $T = 2 \text{ s}$. Give the values of your gains and plot the reference positions of both joints and the actual positions of both joints. You are free to ignore torque limits and friction.

Exercise 11.11 For the two-joint robot of Exercise 11.9 and fifth-order polynomial trajectory of Exercise 11.10, simulate a computed torque controller to stabilize the trajectory. The robot has no joint friction or torque limits. The modeled link masses should be 20% greater than their actual values to create error in the feedforward model. Give the PID gains and plot the reference and

actual joint angles for the computed torque controller as well as for PID control only.

Exercise 11.12 The Krasovskii–LaSalle invariance principle states the following. Consider a system $\dot{x} = f(x)$, $x \in \mathbb{R}^n$ such that $f(0) = 0$ and any energy-like function $V(x)$ such that:

- $V(x) > 0$ for all $x \neq 0$;
- $V(x) \rightarrow \infty$ as $x \rightarrow \infty$;
- $V(0) = \dot{V}(0) = 0$; and
- $\dot{V}(x) \leq 0$ along all trajectories of the system.

Let $\mathcal{S}$ be the largest set of $\mathbb{R}^n$ such that $\dot{V}(x) = 0$ and trajectories beginning in $\mathcal{S}$ remain in $\mathcal{S}$ for all time. Then, if $\mathcal{S}$ contains only the origin, the origin is globally asymptotically stable – all trajectories converge to the origin.

Using the energy function $V(x)$ from Equation (11.40), show how the Krasovskii–LaSalle principle is violated for centralized multi-joint PD setpoint control with gravity compensation if $K_p = 0$ or $K_d = 0$. For a practical robot system, is it possible to use the Krasovskii–LaSalle invariance principle to demonstrate global asymptotic stability even if $K_d = 0$? Explain your answer.

Exercise 11.13 The two-joint robot of Exercise 11.9 can be controlled in task space using the endpoint task coordinates $X = (x, y)$, as shown in Figure 11.25. The task-space velocity is $\mathcal{V} = \dot{X}$. Give the Jacobian $J(\theta)$ and the task-space dynamics model $\{\tilde{\Lambda}(\theta), \tilde{\eta}(\theta, \mathcal{V})\}$ in the computed torque-control law (11.47).

Exercise 11.14 Choose appropriate space and end-effector reference frames $\{s\}$ and $\{b\}$ and express natural and artificial constraints, six each, that achieve the following tasks: (a) opening a cabinet door; (b) turning a screw that advances linearly a distance p for every revolution; and (c) drawing a circle on a chalkboard with a piece of chalk.

Exercise 11.15 Assume that the end-effector of the two-joint robot in Figure 11.25 is constrained to move on the line $x - y = 1$. The robot's link lengths are $L_1 = L_2 = 1$. Write the constraint as $A(\theta)\mathcal{V} = 0$, where $X = (x, y)$ and $\mathcal{V} = \dot{X}$.

Exercise 11.16 Derive the constrained motion equations (11.59) and (11.60). Show all the steps.

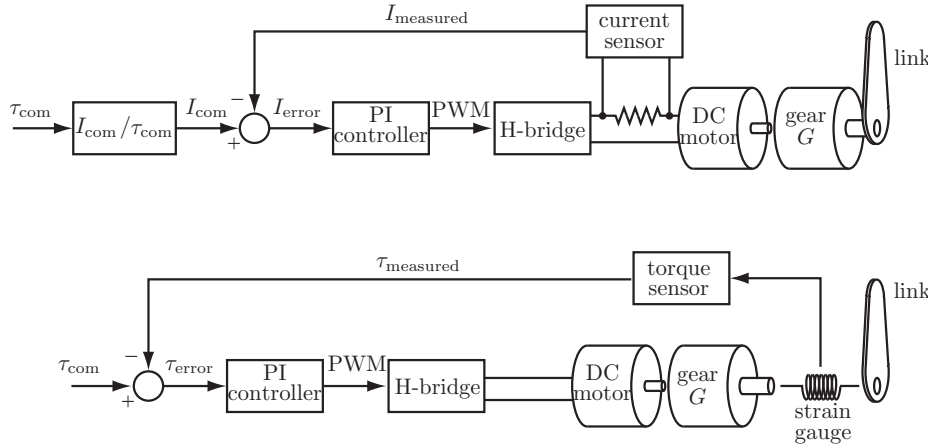


Figure 11.26: Two methods for controlling the torque at a joint driven by a geared DC motor. (Upper) The current to the motor is measured by measuring the voltage across a small resistance in the current path. A PI controller works to make the actual current match better the requested current I_{com} . (Lower) The actual torque delivered to the link is measured by strain gauges.

Exercise 11.17 We have been assuming that each actuator delivers the torque requested by the control law. In fact, there is typically an inner control loop running at each actuator, typically at a higher servo rate than the outer loop, to try to track the torque requested. Figure 11.26 shows two possibilities for a DC electric motor, where the torque τ delivered by the motor is proportional to the current I through the motor, $\tau = k_t I$. The torque from the motor is amplified by the gearhead with gear ratio G .

In the upper control scheme the motor current is measured by a current sensor and compared with the desired current I_{com} ; the error is passed through a PI controller which sets the duty cycle of a low-power pulse-width-modulation (PWM) digital signal and the PWM signal is sent to an H-bridge that generates the actual motor current. In the lower scheme, a strain gauge torque sensor is inserted between the output of the motor gearing and the link, and the measured torque is compared directly with the requested torque τ_{com} . Since a strain gauge measures deflection, the element on which it is mounted must have a finite torsional stiffness. Series elastic actuators are designed to have particularly

flexible torsional elements, so much so that encoders are used to measure the larger deflections. The torque is estimated from the encoder reading and the torsional spring constant.

- (a) For the current sensing scheme, what multiplicative factor should go in the block labeled $I_{\text{com}}/\tau_{\text{com}}$? Even if the PI current controller does its job perfectly ($I_{\text{error}} = 0$) and the torque constant k_t is perfectly known, what effect may contribute to error in the generated torque?
- (b) For the strain gauge measurement method, explain the drawbacks, if any, of having a flexible element between the gearhead and the link.

Exercise 11.18 Modify the `SimulateControl` function to allow initial state errors.